Weekly Research Progress Report

Plant Disease Classification using Transfer Learning | 26 October 2026

1. Objective

Investigate CNN-based plant disease classification using a 15-class PlantVillage subset containing approximately 20,638 images. The current work compares pretrained models as a foundation for later robustness and interpretability experiments.

2. Work completed

• Retained the previously trained EfficientNetV2B0 model as the initial baseline.
• Implemented a lightweight MobileNetV2 variant with ImageNet weights, alpha=0.35, and 160 × 160 feature extraction.
• Froze the pretrained backbone and trained a 15-class classification head with early stopping and checkpointing.
• Saved the best checkpoint and training log in the Kaggle working directory.
• Evaluated the restored model on the existing test dataset using accuracy and macro-averaged metrics.

3. Preliminary MobileNetV2 results

Item	Result
Model	MobileNetV2, alpha=0.35
Feature extraction resolution	160 × 160
Training schedule	30 batches per epoch; early stopping at epoch 7
Validation schedule	10 batches per epoch
Best validation accuracy	83.13% (epoch 5)
Best validation loss	0.6043 (epoch 5)
Test accuracy	81.09%
Macro precision	0.8225
Macro recall	0.7466
Macro F1-score	0.7619

4. Interpretation and limitations

The limited-step MobileNetV2 experiment reached 83.13% validation accuracy and 81.09% test accuracy, with macro F1 of 0.7619. Because only 30 training batches and 10 validation batches were processed per epoch, this is a preliminary experiment, not a fully trained model comparison. Macro metrics should be interpreted alongside per-class results.

The previously recorded EfficientNetV2B0 baseline was approximately 94.75% test accuracy and 0.9458 macro F1. These earlier figures are included as historical baseline values, not as a strictly controlled head-to-head comparison until data partitions, preprocessing, and evaluation are verified as identical.

5. Issues encountered

ConvNeXtTiny training was too slow for the available submission window. CUDA initialization warnings and missing in-memory dataset variables were also encountered. A lightweight MobileNetV2 configuration was used to obtain a preliminary result.

6. Next steps

• Verify fixed, reproducible train/validation/test partitions.
• Train MobileNetV2 with a complete schedule when compute time permits.
• Compare models using consistent preprocessing and metrics.
• Add a confusion matrix, per-class metrics, and Grad-CAM.
• Test robustness to blur, noise, brightness changes, and compression.

Status: Preliminary experiment completed. Controlled training and robustness experiments are required before final conclusions.